

*Robust Real-Time Airway Branch Localization for Bronchoscopy:
Extending BronchoTrack with Anchor Re-Acquisition*

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Dear Sir,

I submit this thesis entitled “Robust Real-Time Airway Branch Localization for Bronchoscopy: Extending BronchoTrack with Anchor Re-Acquisition”, based on MXEN4000 Mechatronic Engineering Research Project 1 and MXEN4004 Mechatronic Engineering Research Project 2, undertaken by me as part-requirement for the degree of B.Eng. in Mechatronic Engineering.

Yours faithfully,

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Abstract

Flexible bronchoscopy relies on the operator's own visual memory and experience to navigate a branching airway tree that offers few distinguishing landmarks, particularly beyond the first few generations of the bronchial tree. Camera-only localization systems, which determine a bronchoscope's position from the video feed alone by comparing detected airway openings against a patient-specific three-dimensional airway model, offer a way to support navigation without the cost, disposable hardware, or procedural complications associated with electromagnetic navigation systems.

This thesis presents BronchoTrack_V3, an implementation and extension of a recently published branch-level bronchoscopic localization method. The baseline system detects airway lumens with a segmentation-based detector, tracks them across frames, and associates each visible lumen with a branch label by comparing its position against a semantic airway graph built from the patient's computed tomography scan. A specific limitation was identified in this baseline: because new detections are only associated relative to a single persistently tracked “anchor” lumen, the system permanently stops reporting a location if that anchor is lost and never recovers, even once clear airway views return. On a 1,084-frame patient bronchoscopy video, this failure mode left the system unable to report a location for approximately 82% of the video following the anchor's loss.

Two candidate solutions were developed and evaluated. The first was a substantial redesign that replaced the anchor entirely with a continuous, per-frame re-evaluation combining three independent signals — a scale-free diameter-growth motion model, point-based bearing matching, and a whole-mask boundary-distance ratio method — fused through a Kalman filter. This redesign was fully implemented and tested but performed worse overall than the existing baseline on the same evaluation video, and was therefore not adopted. The second, adopted solution instead retains the original anchor-based design and adds a bounded, two-stage recovery mechanism: a self re-acquisition check against the last known anchor position, followed by a virtual-anchor fallback that continues to allow new associations without requiring the original lumen to reappear. On the same evaluation video, this increased the proportion of frames with a reported location from 15.3% to 39.9%, a factor of approximately 2.6, while leaving the system's stricter, geometry-verified confidence metric essentially unchanged (14.9% to 14.4%), consistent with the recovery mechanism and the verification check addressing two different questions.

A secondary contribution replaces the system's point-marker visualization with a full segmentation-mask overlay, which improves visual verifiability of detections and incidentally surfaced a minor, pre-existing frame-edge detection artifact. The thesis concludes that a bounded, targeted correction to an already-sound design outperformed a more ambitious redesign for this failure mode, and outlines concrete directions for future work, including relaxing the confidence threshold specifically for recovered labels and automating the CT-to-airway-graph pipeline that currently precedes the system.

Nomenclature

3D	Three-dimensional
CT	Computed tomography
EM	Electromagnetic
EMA	Exponential moving average
EMN	Electromagnetic navigation bronchoscopy
FPS	Frames per second
IoU	Intersection over Union
JSON	JavaScript Object Notation, the file format used for the airway graph and frame logs
MOT	Multi-object tracking
Re-ID	Re-identification (of a tracked object by visual appearance)
SLAM	Simultaneous localization and mapping
SORT	Simple Online and Realtime Tracking
YOLO	You Only Look Once (a family of real-time object detection/segmentation models)

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Chapter 1. Introduction

1.1 Background and Motivation

Flexible bronchoscopy is a minimally invasive procedure in which a thin, camera-tipped instrument is navigated through a patient's airway tree to inspect the airway wall, collect tissue samples, or guide the delivery of a therapeutic device. Unlike many other endoscopic procedures, the airway tree branches repeatedly — from the trachea into the left and right main bronchi, and from there into progressively smaller sub-branches — and offers few visually distinguishing landmarks beyond the first two or three generations of branching. In practice, bronchoscopists navigate largely from memory, training, and an internally held mental model of the patient's airway anatomy built from pre-procedural imaging. Disorientation within this branching structure is a recognized and well-documented challenge, and it disproportionately affects less experienced operators and cases involving anatomically atypical or distorted airways.

A number of commercial and research systems exist to assist with this navigation problem, most notably electromagnetic navigation bronchoscopy (EMN), which tracks a sensor-tipped instrument against a pre-procedural three-dimensional airway map built from the patient's computed tomography (CT) scan. EMN systems are effective but require dedicated sensor hardware, disposable single-use components, specialized capital equipment, and carry a measurable procedural complication rate. This has motivated a separate line of research into camera-only, or vision-based, localization: systems that infer the bronchoscope's position purely from the live video feed, without any additional tracking hardware in the patient or the procedure room.

This thesis is concerned with one such vision-based approach, based on a recently published method (BronchoTrack) that determines the bronchoscope's current airway branch by detecting the openings visible in each video frame, tracking them consistently as the camera moves, and associating each tracked opening with a branch of a pre-built three-dimensional airway graph. This project implements that method as a faithful baseline, identifies a specific and practically significant limitation in it through direct evaluation on real patient video, and develops, evaluates, and compares two candidate solutions to that limitation.

1.2 Problem Statement

The baseline association method identifies a newly visible airway opening only by comparing it against a single, currently trusted, previously identified opening — referred to throughout this thesis as the anchor. This is an effective strategy under normal conditions, since the anchor gives the system a stable reference point from which to reason about which children in the airway graph a newly visible opening could plausibly belong to. However, bronchoscopy video routinely includes brief but severe interruptions: motion blur during rapid instrument movement, the camera briefly pressing against the airway wall, mucus or fluid obscuring the lens, or short stretches in which the detector simply fails to identify any opening at all. If every currently tracked anchor is lost during such an interruption, the baseline system has no mechanism to recover. It permanently stops reporting a location for the remainder of the video, even once a clear, easily identifiable airway view returns moments later.

This is not a theoretical concern. As documented in Chapter 4, evaluating the baseline system on a 1,084-frame patient bronchoscopy video showed the system losing its last anchor roughly one-fifth of the way through the recording and subsequently reporting no location for approximately 82% of the remaining video. A localization system intended to support navigation throughout an entire procedure cannot tolerate a single transient interruption silently ending its usefulness for the remainder of the case.

1.3 Aim and Objectives

The aim of this project is to identify, characterize, and correct this anchor-loss failure mode in a working bronchoscopic localization system, without compromising the accuracy the baseline system already achieves under normal conditions. This aim is addressed through the following objectives:

- Implement a faithful baseline reproduction of the published BronchoTrack method, including lumen detection, multi-object tracking, and graph-based airway association, as a reference against which any modification can be directly compared.
- Characterize the anchor-loss limitation quantitatively on real patient video, rather than by inspection of the method alone.
- Design, implement, and evaluate a substantial alternative redesign that removes the anchor dependency entirely, to test whether a more ambitious architectural change is warranted.
- Design, implement, and evaluate a smaller, targeted recovery mechanism that retains the original anchor-based design, as a more conservative alternative to the redesign.
- Compare both candidates directly on the same evaluation video and adopt the better-performing approach as the recommended, maintained system.
- Improve the system's output visualization from a single point marker to a full detected-shape overlay, to make the system's behaviour easier to verify visually.

1.4 Scope

This project is concerned with branch-level localization — determining which named airway branch the bronchoscope currently occupies — rather than continuous six-degree-of-freedom pose estimation within a branch. The airway graph itself is assumed to already exist as a labeled three-dimensional structure derived from a prior CT scan; the process of segmenting a CT volume and extracting that graph is treated as a prerequisite input to this system rather than a contribution of this thesis, though its automation is discussed as future work in Chapter 6. Evaluation in this project is conducted on recorded patient bronchoscopy video rather than live, in-procedure operation, consistent with the retrospective evaluation methodology used in the published baseline method this project builds on.

1.5 Thesis Structure

Chapter 2 reviews existing approaches to bronchoscopic navigation, situates the baseline method this project extends within that literature, and identifies the specific

gap this thesis addresses. Chapter 3 describes the system architecture, the materials and data used, and the methodology behind both candidate solutions to the anchor-loss problem, together with the evaluation methodology used to compare them. Chapter 4 presents the quantitative results of that evaluation and discusses their implications, including the results of the visualization improvement. Chapter 5 draws conclusions from this work, and Chapter 6 outlines directions for future work.

Chapter 2. Background and Literature Review

Approaches to assisting bronchoscopic navigation broadly fall into three categories: electromagnetic and other external-sensor tracking, image-registration methods that align the live video with a pre-built three-dimensional model (sometimes described as virtual bronchoscopy or visual SLAM), and detection-and-association methods that reason about discrete anatomical landmarks — principally airway bifurcations — visible in the video itself. This chapter reviews representative work in each category before introducing the specific method this thesis builds on.

2.1 Electromagnetic Navigation and Robotic-Assisted Bronchoscopy

Electromagnetic navigation bronchoscopy (EMN) is the most clinically established approach to bronchoscopic guidance. A thin-slice (around 1 mm) CT scan of the chest, typically acquired at full inspiration, is used to build a virtual bronchoscopic model of the airway tree, segment a target lesion, and plan a pathway from the central airway to that lesion. During the procedure, a sensor probe passed through the bronchoscope's working channel is tracked within an externally generated electromagnetic field and registered against this pre-built map, allowing the operator to follow the planned pathway to the target [1]. Two platforms dominate current clinical use, superDimension (Medtronic) and the SPiN system (Olympus) [1], [8].

Reported diagnostic yield for EMN varies widely across the literature, from as low as 33% to as high as 88% [1], a range Lin, Ho, and Frye attribute largely to an early evidence base of small, single-centre studies [8]. The NAVIGATE trial, the largest prospective multi-centre evaluation of EMN to date, offers a more generalizable figure: its one-year U.S. cohort of 1,157 patients reported a 73% diagnostic yield and 69% sensitivity for malignancy at a median lesion size of 20 mm, with a 2.9% pneumothorax rate, while a subsequent two-year follow-up expanding to 1,388 patients across the United States and Europe reported a somewhat lower 67.8% yield, 62.6% sensitivity, and 4.7% pneumothorax rate. A positive radiographic “bronchus sign” — an airway visibly leading into the lesion on the planning CT — was present in only about half of cases, and was, alongside operator experience and the use of rapid on-site cytological evaluation, one of the strongest predictors of diagnostic success [8].

Even at this validated level, EMN's yield still falls short of CT-guided transthoracic needle aspiration, and at least one meta-analysis found no clear advantage over the substantially cheaper combination of conventional bronchoscopy with radial endobronchial ultrasound (r-EBUS) [8]. The leading technical explanation for this gap is CT-to-body divergence: the planning CT is acquired at full lung inflation, while the procedure itself is performed under general anaesthesia and positive-pressure ventilation nearer functional residual capacity, so the lesion's true position at the time of sampling can differ from its planned position by several centimetres — an effect reported to be more pronounced for lower-lobe lesions and to have shifted lesions by as much as 4 cm in at least one study [8]. r-EBUS is frequently added specifically to confirm real-time lesion position against this drift, though it depends on a positive bronchus sign to be useful at all — one study found diagnostic yield fell from 79% to 31% without one — and has itself been reported to give false-positive confirmation

against atelectatic tissue that ultrasonically resembles the lesion [8]. A further practical concern specific to EMN is the theoretical risk of electromagnetic interference with implanted cardiac devices, though the small safety studies conducted to date have not observed any actual disruption to device function [8].

Robotic-assisted bronchoscopy (RAB) is a more recent, related approach that keeps the same CT-planned-pathway strategy while adding continuous robotic control, intended to improve reach and stability at the periphery of the lung where a conventional, EMN-guided bronchoscope becomes difficult to steer precisely [8]. Three platforms have reached clinical use: the Monarch system (Auris Health), controlled with a handheld interface and tracked electromagnetically, which wedges a 6.0 mm outer sheath at the segmental airways and advances a 4.4 mm inner scope the rest of the way to the target; the Ion platform (Intuitive Surgical), controlled with a trackball and wheel and tracked instead through fibre-optic shape sensing along a 3.5 mm fully articulating catheter, avoiding an external electromagnetic field entirely; and the Galaxy system (Noah Medical), a disposable, EMN-guided bronchoscope that adds continuous digital tomosynthesis and augmented fluoroscopy so the operator retains real-time visual confirmation throughout sampling itself, rather than relying solely on the pre-planned route [8].

For the purposes of this thesis, the significance of both EMN and RAB is less their individual diagnostic performance than what they hold in common: whatever their differences in tracking modality, both remain dependent on dedicated capital equipment, proprietary single-use catheters or sheaths, and, for the electromagnetically tracked platforms, an external field generator. RAB improves on EMN's reach and stability, but does not remove the cost and hardware burden that motivates camera-only localization in the first place (Section 1.1) — it relocates that burden onto a more capable, but still equally hardware-dependent, platform. This is the gap the vision-based methods reviewed in the remainder of this chapter are aimed at.

2.2 Image-Registration and Landmark-Based Vision Methods

A separate line of work uses the bronchoscope's video feed itself, without external sensors, either by registering the live view against a virtual rendering of the CT-derived airway model (an approach sometimes framed as visual simultaneous localization and mapping, or visual SLAM) or by detecting specific anatomical landmarks — principally the bifurcations where one airway branch splits into two or more children — and reasoning about the instrument's position relative to those landmarks.

Shen, Giannarou, Shah, and Yang's BRANCH method is representative of the landmark-based approach: it constructs a shape-context airway descriptor specifically to detect and characterize bifurcations in the distal airway, aiming to support localization deeper into the bronchial tree than image-appearance matching alone typically allows [3]. This reflects a broader theme in the literature: as the airway tree branches into smaller and more visually similar sub-branches, appearance-based registration against the CT-derived model becomes progressively less reliable, and structural landmarks — specifically, the openings that lead to each branch — become a more robust cue than overall scene appearance.

This same insight underlies the detection-and-association family of methods this thesis is most directly built on, which treats each airway opening as a discrete, trackable object rather than registering the whole visual scene, and is described in the following section.

2.3 Detection-and-Association Methods and the BronchoTrack Baseline

Tian, Liao, Huang, Yang, Wu, Chen, Li, and Liu's BronchoTrack method — the method this thesis implements and extends — frames branch-level localization as a three-stage pipeline: a lightweight detector identifies airway lumens (openings) visible in each frame, a multi-object tracking stage maintains each lumen's identity consistently across frames despite rapid camera motion, and a training-free association stage matches each currently tracked lumen against a semantic airway graph that encodes the known hierarchy and branching structure of the patient's own airway tree [4]. This avoids the need to register the full visual scene against a rendered CT model on every frame, and instead reduces the problem to matching a small number of discrete, well-defined openings. The original work reports 85.64% localization accuracy across nine patient datasets up to the fourth airway generation, and successful navigation to the eighth generation in porcine in-vivo testing [4].

The same research group's subsequent PANS system reformulates branch-level localization probabilistically, aiming for improved robustness under the kind of transient visual ambiguity and interruption this thesis is specifically concerned with [5]. This is a useful point of comparison for the present work: rather than adopting a full probabilistic reformulation of the underlying association method, this thesis instead evaluates whether a smaller, more targeted correction to the existing anchor-based design — confined specifically to the anchor-loss failure mode — can achieve a comparable practical improvement without the added complexity of a broader redesign.

The multi-object tracking stage in this family of methods draws on general-purpose tracking techniques developed outside the medical imaging literature. Bewley, Ge, Ott, Ramos, and Upcroft's SORT (Simple Online and Realtime Tracking) algorithm — which associates detections across frames using a Kalman filter for motion prediction combined with the Hungarian algorithm for optimal assignment — established the basic motion-based matching pattern that this project's tracking stage, and BronchoTrack's original tracking stage, both follow [6]. The lumen detector itself is a real-time object detection and segmentation network from the YOLO (You Only Look Once) family, originally introduced by Redmon, Divvala, Girshick, and Farhadi as a single-pass alternative to slower region-proposal-based detectors [7], and realized in this project using the YOLO26 architecture described by Sapkota, Cheppally, Sharda, and Karkee [8].

2.4 Gap Addressed by This Thesis

The methods reviewed above establish detection-and-association against a semantic airway graph as an effective, hardware-free approach to branch-level bronchoscopic localization, with BronchoTrack specifically as the direct basis for this project's implementation. However, published evaluations of this family of methods report

overall accuracy figures without isolating the specific behaviour of the system during and immediately after a total tracking interruption — a scenario this thesis found, through direct evaluation on real patient video, to be a severe and specific failure mode of the anchor-dependent association design. This thesis addresses that gap directly: it characterizes the anchor-loss failure mode quantitatively, evaluates a substantial redesign intended to eliminate it entirely, and evaluates a smaller, targeted recovery mechanism as a more conservative alternative, reporting an honest, like-for-like comparison between the two on the same evaluation data.

Chapter 3. Experimental Procedure

3.1 System Architecture

The implemented system processes bronchoscopy video one frame at a time through four sequential stages, illustrated in Figure 1: lumen detection identifies the airway openings visible in the current frame; multi-lumen tracking maintains each opening's identity consistently from one frame to the next; airway association determines which branch of the patient's airway graph each currently visible, tracked opening most likely corresponds to; and localization takes a vote among all currently labeled openings to report the bronchoscope's overall current branch. Two design choices in this architecture are significant enough to warrant discussion before the individual stages are described in detail.

First, the detection stage produces a full segmentation mask for each detected opening, not merely a bounding box. This is a deliberate choice: an opening's shape and apparent size carry useful evidence for identifying which branch it is (a larger, rounder opening close to the camera reads differently from a small, foreshortened one further away), and this information would be discarded under a box-only detector. In the implemented system, this mask information is used specifically by the association stage's diameter-based verification step, described in Section 3.4; the tracking stage itself remains deliberately box-based, following the established motion-and-appearance matching pattern used by general-purpose trackers such as SORT [6], since full mask-to-mask matching on every frame would add tracking-stage cost without a demonstrated accuracy benefit for this system's design. This trade-off is revisited as a future-work item in Chapter 6.

Second, the system performs an independent verification step after association: once an opening is matched to a branch by position, its apparent size at the camera's estimated distance from that branch is separately checked against the size the airway graph predicts for that branch at that distance. A label is only treated as fully confirmed — and rendered as a confirmed detection in the output visualization — once both the positional match and this size check agree. This is a stricter standard than it may initially appear, and it becomes directly relevant to interpreting the results in Chapter 4, since this verification metric behaves differently from the overall labeling rate reported alongside it.

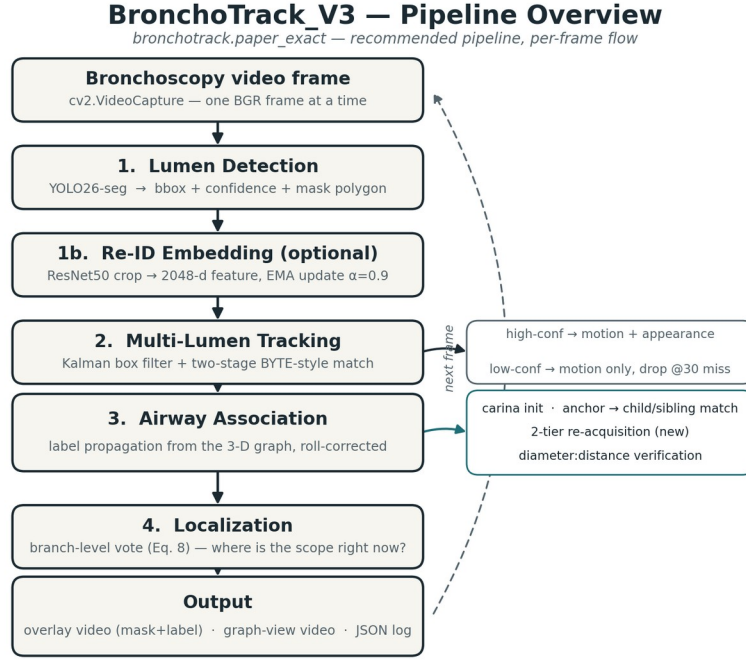


Figure 1. System architecture: how a single video frame moves through the implemented pipeline, from input to output.

3.2 Materials

Three inputs are required by the system. The first is bronchoscopy video; the evaluation reported in Chapter 4 uses a 1,084-frame recording of a real patient bronchoscopy procedure. The second is a patient-specific airway graph: a labeled, hierarchical three-dimensional representation of the patient's own airway tree, derived in advance from their CT scan and encoding, for each named branch, its parent branch, its generation (depth from the trachea), its centerline, and its physical radius at points along that centerline. This graph is treated in this project as a pre-existing input artifact rather than a contribution of this thesis; its own construction pipeline, which uses surface-mesh network extraction to convert a segmented CT-derived mesh into this labeled graph structure, is summarized only briefly here and revisited as a candidate for automation in Chapter 6. The third input is the trained lumen detector itself: a YOLO26 segmentation model [8], fine-tuned to detect airway lumens and output, for each detection, a bounding box, a confidence score, and a segmentation mask polygon outlining the opening's true shape.

3.3 Detection and Tracking

Each frame is passed through the YOLO26 segmentation detector, which returns zero or more detections above a configured confidence threshold. Each detection carries a bounding box, confidence score, and mask polygon. These detections are passed to a multi-object tracking stage responsible for maintaining a consistent identity for each physical airway opening across frames, so that the same opening is not mistaken for a newly appearing one purely because the camera moved between frames.

Tracking follows a Kalman-filter-based motion model over each tracked object's bounding box (center position, height, and aspect ratio), predicting each tracklet's expected position in the current frame before matching it against the current frame's detections — the same basic pattern established by SORT [6]. Matching proceeds in two tiers, following a BYTE-style strategy: high-confidence detections are matched against existing tracklets using a combination of motion (predicted position overlap) and appearance similarity, drawn from a ResNet50 feature embedding of each detection's image crop, updated over time with an exponential moving average to remain robust to gradual appearance change; low-confidence detections are matched using motion alone, which allows the tracker to continue following an opening even through a few frames of degraded detection confidence without immediately discarding it. A tracklet that is not matched to any detection for 30 consecutive frames is dropped.

3.4 Baseline Airway Association

The association stage is responsible for assigning each currently tracked lumen a branch label from the airway graph, and is where this project's core contribution is situated. The baseline design, following the published BronchoTrack method [4], operates around a single trusted reference detection at a time, referred to throughout this thesis as the anchor: the tracklet most recently and confidently identified as occupying the airway branch the bronchoscope currently sits in.

Association proceeds in three steps. First, at the very start of a video, before any anchor exists, the system bootstraps by matching the visible lumens at the first bifurcation (the carina, where the trachea splits into the left and right main bronchi) directly against the graph's known geometry, since no prior reference point exists yet to match relative to. Second, for every subsequent frame, each newly visible, currently unlabeled lumen is compared against the anchor's known position and the graph's predicted layout of the anchor's own child and sibling branches, correcting for the camera's estimated roll angle about its own optical axis, to determine which child branch it most plausibly corresponds to. Third, once a candidate label passes this positional matching step, it is subjected to the diameter:distance verification described in Section 3.1: the candidate's apparent size in the frame is compared against the size the graph predicts for that branch at the camera's current estimated distance, and the label is only treated as fully confirmed if both checks agree.

This anchor-centred design is effective under normal conditions, since it reduces every association decision to a comparison against one well-understood, currently visible reference point rather than an unconstrained search over the whole graph. Its limitation, characterized quantitatively in Chapter 4, is that it has no defined behaviour for the case where the anchor itself is lost — through motion blur, occlusion, or a period of detector failure — and no other opening has yet been confidently identified to replace it.

3.5 Candidate Solution A: Continuous Multi-Signal Re-Evaluation

The first candidate solution investigated was a substantial redesign that removes the anchor concept entirely. Rather than persistently depending on any single tracked

object, this design re-evaluates every currently visible lumen against every plausible candidate branch on every single frame, combining three independent signals into one fused confidence score per candidate branch.

The first signal is a diameter-growth motion model: each tracklet's own apparent diameter is tracked over time with a Kalman filter, and its smoothed rate of change is used to derive a scale-free time-to-contact estimate — the number of frames until the camera reaches that opening — without requiring any camera calibration, since the estimate depends only on the ratio of current diameter to its own rate of growth. These implied time-to-contact values are cross-checked, though never used to calibrate the model itself, against the median closing speed observed across the video's own already-completed branch transitions, flagging candidates whose implied approach speed is wildly inconsistent with that history as implausible. The second signal is a direct re-derivation of the baseline's own positional matching idea: comparing each tracklet's segmentation-mask center against the graph's predicted bearing to each candidate child branch. The third signal, intended to make genuine use of the detector's mask output rather than reducing it to a single point, compares each tracklet's whole mask shape against the graph's predicted branch geometry, using an area-equivalent diameter computed from the full mask boundary and the nearest boundary-to-boundary gap between two mask polygons, rather than a simple center-to-center distance.

These three signals are combined per candidate branch through a Hungarian assignment followed by a per-branch Kalman filter that smooths the fused confidence over time, and a branch transition is only committed once a candidate's smoothed confidence clears a threshold for a run of consecutive frames, rather than on a single strong frame. This design was fully implemented, covered by an automated test suite (Section 3.7), and evaluated on the same 1,084-frame video used to characterize the baseline's anchor-loss limitation.

3.6 Candidate Solution B: Bounded Anchor Recovery

The second, more conservative candidate solution retains the baseline's original anchor-centred design in full, and adds a bounded recovery mechanism specifically for the case where every anchor is lost simultaneously. The system retains a snapshot of the most recently confirmed anchor — its position, its branch label, and the frame at which it was last confidently observed — even after that opening is no longer visible or trackable.

If every anchor is lost, recovery proceeds in two stages. First, the system checks whether any currently unlabeled opening is likely the same physical opening reappearing — that is, the bronchoscope's own current position, returning into view after a brief interruption — by measuring how closely its position overlaps the anchor snapshot's last known position (intersection-over-union against that frozen reference). If the overlap exceeds a threshold, the opening is relabeled directly with the anchor's own former label, without requiring a fresh graph-based match. Second, if no sufficiently close match is found, the system falls back to treating the anchor snapshot's last known position as a virtual reference point, allowing newly visible openings

nearby to still be matched against the graph's predicted layout of the anchor's children and siblings, even though the original physical opening itself has not reappeared.

In either case, once an opening is successfully re-identified through this mechanism, the system resumes normal anchor-based operation immediately and treats the newly identified opening as the current anchor; this is not a one-time reset, and the mechanism can recur as many times as needed over the course of a video. A safeguard is included to avoid acting on a stale reference: if more than approximately three seconds have elapsed since the retained anchor snapshot was itself confidently observed, the system treats it as unreliable and withholds recovery until a new anchor is established through the normal bootstrap or matching process, rather than propagating an outdated position indefinitely.

3.7 Evaluation Methodology

Both candidate solutions were evaluated on the same 1,084-frame real patient bronchoscopy video used to originally characterize the baseline's anchor-loss limitation, under otherwise identical detector, tracker, and graph configuration, to allow a direct, like-for-like comparison. Two metrics are reported for each run: the proportion of frames for which the system reports any location at all (reflecting whether the anchor-loss failure mode has been addressed), and the proportion of frames for which the stricter, geometry-verified diameter:distance check described in Section 3.1 is additionally satisfied (reflecting whether label confidence itself has been affected, independent of the recovery behaviour). The quantitative results of this comparison, and the resulting decision to adopt Candidate Solution B over Candidate Solution A, are reported in Chapter 4.

In addition to this evaluation-video comparison, the implementation includes an automated regression test suite covering the individual software components of the system, developed and maintained throughout the project as each stage was implemented. This suite comprises 58 tests across three modules — the general pipeline, the baseline association logic including the recovery mechanism, and the Candidate Solution A redesign — summarized in Table 1, and is used during development to catch regressions in individual components; it is distinct from, and does not substitute for, the end-to-end evaluation-video comparison reported in Chapter 4, which was carried out directly against real patient video with the recovery mechanism enabled and disabled under otherwise identical settings.

Module	Tests
General pipeline	17
Baseline association (incl. Candidate Solution B recovery)	23
Candidate Solution A (multi-signal redesign)	18
Total	58

Table 1. Automated regression test suite, by module.

3.8 Visualization

The system's output video originally marked each identified lumen with a small dot at its geometric center. As a secondary improvement, this was replaced with a rendering

of the detector's actual segmentation mask as a filled, translucent, outlined overlay, gated on the diameter:distance verification check described in Section 3.1 so that only fully confirmed detections receive the full overlay treatment, with a fallback to the original dot marker where no usable mask is available. This change and its incidental findings are discussed in Chapter 4.

Chapter 4. Results and Discussion

4.1 Baseline Anchor-Loss Characterization

Running the unmodified baseline system on the 1,084-frame evaluation video confirmed the failure mode motivating this thesis. The system lost its last active anchor roughly one-fifth of the way through the recording — during a stretch of rapid camera motion and partial occlusion — and, having no mechanism to recover, reported no location for approximately 82% of the remaining video, despite the video subsequently returning to clear, easily identifiable airway views on multiple occasions. This result, on its own, establishes that the anchor-loss limitation identified in Chapter 1 is not an edge case but a severe and practically significant failure mode for this class of system.

4.2 Comparison of Candidate Solutions

Candidate Solution A (continuous multi-signal re-evaluation) was fully implemented and evaluated on the same video. Upon direct comparison against the baseline, it performed worse overall: the added complexity of continuously re-deriving and fusing three independent signals every frame did not translate into a corresponding improvement in reliability, and in practice introduced its own new sources of noise, particularly in the motion model's diameter-growth estimate during frames where a tracked opening's apparent size briefly fluctuated for reasons unrelated to genuine approach or recession. Based on this direct comparison, Candidate Solution A was set aside rather than adopted.

This negative result is nonetheless a meaningful finding in its own right, and is reported here rather than omitted. It indicates that the baseline's core anchor-based association strategy was fundamentally sound, and that the anchor-loss problem specifically required a bounded, targeted correction rather than a wholesale architectural replacement. Candidate Solution A remains in the project's codebase, fully implemented and tested, should a future extension of this work find a specific use for its continuous re-evaluation design — for instance, a scenario with a much noisier detector where no single anchor can be trusted for long — that this project's own evaluation video did not present.

Candidate Solution B (bounded anchor recovery) was evaluated on the same video under otherwise identical settings, comparing a run with the recovery mechanism disabled against one with it enabled. Table 2 reports the results of this comparison.

Metric	Before	After
Frames with a reported location	15.3%	39.9%
Frames with a fully confirmed (geometry-verified) detection	14.9%	14.4%

Table 2. Same 1,084-frame evaluation video and settings, Candidate Solution B, before and after the recovery mechanism.

The proportion of frames with a reported location increased by a factor of approximately 2.6, directly confirming that the system no longer permanently stops reporting a location following a total anchor-loss interruption of the kind characterized

in Section 4.1. Unlike Candidate Solution A, this improvement was achieved without a corresponding regression in overall reliability, since the recovery mechanism only activates in the specific circumstance where no anchor currently exists, and leaves the system's normal anchor-based matching behaviour, including the diameter:distance verification step, entirely unchanged the rest of the time.

4.3 The Confirmed-Detection Metric

One result in Table 2 warrants direct discussion: the stricter, geometry-verified confidence metric remained essentially unchanged before and after the recovery mechanism was enabled (14.9% to 14.4%). This is not an inconsistency, and it does not indicate that the recovery mechanism failed to work as intended. The recovery mechanism and the diameter:distance verification check answer two different questions: the recovery mechanism determines whether the system can resume reporting a location at all following an interruption, while the verification check separately determines how strictly the system should trust any given match, independent of how that match came to exist. Because recovery restores a label through the same position-matching logic used everywhere else in the baseline design, and does not relax the subsequent verification step in any way, a recovered label is held to exactly the same strict standard as one identified under normal conditions. Whether this verification threshold should be deliberately relaxed specifically for labels restored through recovery, given that a recovered anchor is by definition working from a position estimate briefly out of date, is identified as an open design question and is discussed further as future work in Chapter 6.

4.4 Visualization Improvement

Replacing the point-marker output with a full segmentation-mask overlay, as described in Section 3.8, considerably improved the ability to visually verify what the system is detecting in the output video, since a viewer can directly compare the rendered mask shape against the visible airway opening rather than inferring correctness from a single point. This change also surfaced a minor, pre-existing detection artifact that had not been visually obvious under the previous point-marker rendering: an opening located near the edge of the video frame is occasionally detected with an irregular, oversized mask, because the detector's mask boundary follows the edge of the visible frame rather than terminating cleanly at the true physical edge of the opening. This is a property of the raw detection output itself rather than a defect introduced by the tracking or association stages, and is noted as a candidate for a dedicated filtering step in future work (Chapter 6) should it prove distracting in practice.

4.5 Summary

Taken together, these results support adopting Candidate Solution B — the bounded, two-stage anchor recovery mechanism — as the system's recommended, maintained design. It directly and substantially addresses the anchor-loss failure mode characterized in Section 4.1, without the reliability regression observed in the more ambitious Candidate Solution A redesign, and without disturbing the baseline's existing, already-effective normal-condition behaviour. The one open question this evaluation surfaces — whether the confirmed-detection verification threshold should

differ for recovered labels specifically — is a refinement to an already-working mechanism rather than evidence against the mechanism itself, and is carried forward as future work.

Chapter 5. Conclusions

This thesis set out to identify and correct a specific limitation in a published vision-based bronchoscopic localization method: the total and permanent loss of location reporting following any interruption severe enough to lose every currently tracked anchor lumen simultaneously. This limitation was first confirmed quantitatively rather than assumed, by evaluating a faithful baseline reproduction of the published method on a real, 1,084-frame patient bronchoscopy video, where it left the system unable to report a location for approximately 82% of the video following an anchor-loss event roughly one-fifth of the way through the recording.

Two candidate solutions were developed, implemented, and evaluated under identical conditions. The more substantial redesign — replacing the anchor concept entirely with a continuous, multi-signal per-frame re-evaluation — was found, on direct comparison, to perform worse overall than the existing baseline, and was not adopted. The smaller, more conservative solution — retaining the original anchor-based design and adding a bounded, two-stage recovery mechanism triggered only when every anchor is simultaneously lost — increased the proportion of the evaluation video with a reported location by a factor of approximately 2.6, while leaving the system's independent, stricter geometry-verification metric essentially unaffected, consistent with the two mechanisms addressing genuinely separate questions. This solution was adopted as the system's recommended, actively maintained design.

A secondary improvement, replacing the system's point-marker output visualization with a full segmentation-mask overlay, measurably improved the visual verifiability of the system's detections and, as a direct consequence of that improved verifiability, surfaced a minor pre-existing detection artifact at the video frame's edge that had not previously been visually apparent.

The central methodological conclusion of this work is that, for the specific failure mode investigated here, a bounded and targeted correction to an already-sound design outperformed a more ambitious architectural redesign intended to eliminate the same failure mode by construction. This is a useful and generalizable finding for future work on this and related systems: it demonstrates that the anchor-based association strategy underlying BronchoTrack-style methods is fundamentally robust, and that isolated, well-characterized failure modes within such a design are often better addressed with equally isolated corrections than with wholesale replacement.

Chapter 6. Future Work

Several concrete directions for future work follow directly from the results and open questions identified in Chapter 4.

6.1 Confidence Threshold for Recovered Labels

As discussed in Section 4.3, the recovery mechanism restores a location label without relaxing the subsequent diameter:distance verification check that governs whether a label is displayed as fully confirmed. Since a recovered anchor is, by definition, working from a position estimate that may be briefly out of date, future work should investigate whether a deliberately more lenient verification threshold — applied specifically to labels restored through recovery, rather than globally — would improve the confirmed-detection rate during and immediately after a recovery event without compromising overall label reliability.

6.2 Frame-Edge Detection Artifact

The frame-edge mask distortion artifact surfaced by the visualization improvement in Section 4.4 was not directly addressed in this project, since it is a property of the raw detector output rather than of the tracking or association stages this thesis is centred on. A dedicated filtering or post-processing step — for instance, suppressing or flagging masks whose boundary closely follows the video frame's own edge — is a well-scoped and low-risk piece of future work that would further improve output quality.

6.3 Automating the CT-to-Airway-Graph Pipeline

This project treats the patient-specific, CT-derived airway graph as a pre-existing input, consistent with the scope defined in Chapter 1. Automating the full pipeline from a raw CT scan through airway segmentation, three-dimensional surface mesh reconstruction, and semantic graph extraction into a single, streamlined preprocessing step would make the overall system substantially more practical to deploy against new patients, and is a natural, self-contained extension of this work.

6.4 Re-examining Candidate Solution A in a Noisier Setting

As discussed in Section 4.2, Candidate Solution A's continuous multi-signal re-evaluation design was not adopted because it underperformed the simpler baseline on this project's evaluation video. This does not rule out the design being genuinely useful in a different operating regime — for instance, against a substantially noisier detector, a lower frame rate, or an airway anatomy with unusually ambiguous branch geometry, where no single anchor could plausibly be trusted for an extended period. Future work could evaluate Candidate Solution A specifically under such conditions, rather than against the same evaluation video used throughout this thesis, before concluding more generally on its usefulness.

6.5 Broader Clinical Evaluation

All results in this thesis are drawn from a single 1,084-frame patient bronchoscopy video. A broader evaluation across multiple patients, airway anatomies, and bronchoscopy equipment would be needed before any conclusion here could be

considered clinically generalizable, and is the natural next step beyond the scope of this thesis.

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Appendix A: Candidate Solution A — Implementation Reference (Pseudocode)

This appendix gives the exact algorithmic detail behind Candidate Solution A, the continuous multi-signal re-evaluation redesign discussed in Sections 3.5 and 4.2, which was fully implemented and evaluated but not adopted as the recommended system. It is included here as a complete reference, since it remains in the project codebase and represents a meaningful part of the development and evaluation carried out for this thesis. Each entry below gives one source file's role in a line, followed by a functional walkthrough of its classes and functions in the order they execute.

bronchotrack/scalar_kalman.py

Shared 2-state constant-velocity Kalman filter ([value, rate]), reused by both paper_exact (roll angle, diameter:distance smoothing) and fusion (diameter growth rate, per-label confidence smoothing).

```
function build_matrices(dt):
    F = [[1, dt], [0, 1]]      # state transition: value += rate*dt
    H = [[1, 0]]              # we only observe "value", not "rate"
    return F, H

class ScalarKalmanFilter:
    state: [value, rate], covariance P, process_noise Q, measurement_noise R

    function predict(dt):
        F, _ = build_matrices(dt)
        state = F . state
        P = F . P . F_transpose + Q
        return self

    function update(measurement):
        F, H = build_matrices(dt=1)
        y = measurement - H . state          # innovation
        S = H . P . H_transpose + R
        K = P . H_transpose . inverse(S)      # Kalman gain
        state = state + K . y
        P = (I - K . H) . P
        return self

    property value: state[0]
    property rate: state[1]
```

bronchotrack/fusion/motion_model.py

Diameter-growth motion model: tracks each tracklet's apparent diameter (Kalman-smoothed) and derives a scale-free time-to-contact. Cross-checked (never calibrated) against the airway graph's known branch lengths.

```
class ApproachEstimate:
    fields: track_id, diameter, growth_rate, approaching (bool), tau_frames (nullable)

    function detection_diameter(tracklet):
        return tracklet's box height      # proxy for apparent lumen size

class ApproachMotionModel:
    filters: map of track_id -> ScalarKalmanFilter    # smooths diameter + its rate of change

    function update(tracklets, frame_idx):
```

```

    estimates = {}
    for t in tracklets:
        diam = detection_diameter(t)
        filt = filters.get(t.track_id) or new ScalarKalmanFilter(seeded at diam)
        filt.predict(dt=1); filt.update(diam)
        filters[t.track_id] = filt

        growth_rate = filt.rate
        approaching = growth_rate > small_epsilon      # only "approaching" if
genuinely growing
        tau_frames = filt.value / growth_rate if approaching else None
        # tau = diameter / d(diameter)/dt -- scale-free time-to-contact
        estimates[t.track_id] = ApproachEstimate(t.track_id, filt.value,
growth_rate,
                                                approaching, tau_frames)

    return estimates

function tau_frames_for(track_id):
    return estimates.get(track_id).tau_frames if present else None

function prune(live_track_ids):
    remove any entry in filters whose track_id is not in live_track_ids

class TransitRecord:
    fields: label, frames_spent, arc_length_mm

class TransitSanityChecker:
    plausible_range: (low, high)      # mm per frame bounds
    history: list of TransitRecord

    function record_transit(label, frames_spent, arc_length_mm):
        append TransitRecord to history

    property median_speed_mm_per_frame:
        return median(arc_length_mm / frames_spent for each record), or None if empty

    function check(tau_frames, candidate_arc_length_mm):
        if history is empty: return "unknown"
        implied_speed = candidate_arc_length_mm / tau_frames
        ratio = implied_speed / median_speed_mm_per_frame
        return "plausible" if plausible_range.low <= ratio <= plausible_range.high
        else "implausible"      # empirical, self-referential check, not a
calibration

```

bronchotrack/fusion/bronchotrack_id.py

'The bronchotrack version': point-based bearing matching using each lumen's segmentation center, re-derived fresh rather than importing paper_exact.association's implementation.

```

function detection_point(tracklet):
    return polygon_opposing_center(tracklet.mask) if tracklet has a mask
    else tracklet's box center

function bearing_cost_matrix(graph, ref_label, candidate_labels, detections,
reference_point):
    if candidate_labels is empty: return None
    projected = graph.project_children_2d(ref_label)    # 2D unit bearing per candidate
child
    if any candidate label is missing a projection: return None

```

```

cost = matrix[len(candidate_labels)][len(detections)]
for i, label in candidate_labels:
    for j, det in detections:
        observed_bearing = normalize(detection_point(det) - reference_point)
        cost[i][j] = 1 - cosine_similarity(observed_bearing, projected[label])
return cost      # 0 = perfect bearing match, up to 2 = opposite direction

```

bronchotrack/fusion/ratio_id.py

'The ratio method': genuinely whole-mask, not point-based. Size is the mask's own area-equivalent diameter; distance is the nearest boundary-to-boundary gap between two full mask polygons.

```

function mask_boundary_gap(poly_a, poly_b):
    if either polygon is None: return None
    return min distance over all point pairs (p in poly_a, q in poly_b)
    # nearest boundary-to-boundary gap, an all-pairs search over both boundaries

function circle_boundary_gap(center_a, radius_a, center_b, radius_b):
    d = distance(center_a, center_b) - radius_a - radius_b
    return max(d, MIN_DIST)    # clamp so overlapping circles never go to zero/negative

function reference_opening_radius(graph, ref_label):
    return graph node ref_label's known physical radius, or None if graph has no radius data

function ratio_cost_matrix(graph, ref_label, candidate_labels, reference_detection,
                           detections, virtual_advance_mm):
    ref_radius = reference_opening_radius(graph, ref_label)
    if ref_radius is None or detections is empty or reference_detection.mask is None:
        return None

    ref_equiv_diameter = mask_equivalent_diameter(reference_detection.mask)
    # area-based (shoelace integral over the mask boundary), not longest-diameter
    px_per_mm = ref_equiv_diameter / (2 * ref_radius)

    cost = matrix[len(candidate_labels)][len(detections)]
    for i, label in candidate_labels:
        child_radius = graph.child_apparent_equivalent_diameter_at_distance(
            ref_label, label, virtual_advance_mm) / 2
        for j, det in detections:
            if det.mask is None: cost[i][j] = large_penalty; continue
            gap_px = mask_boundary_gap(reference_detection.mask, det.mask)
            gap_mm = gap_px / px_per_mm
            expected_gap_mm = circle_boundary_gap(using ref_radius and child_radius)
            cost[i][j] = abs(gap_mm - expected_gap_mm)
            # how well this detection's real-world spacing matches the graph's
            expectation
    return cost

```

bronchotrack/fusion/kalman_fusion.py

Blends the bearing and ratio cost matrices into one Hungarian assignment per frame, nudges the result by the motion model, and smooths each candidate's confidence with a per-label Kalman filter.

```

class CandidateConfidence:
    fields: label, fused_confidence, matched_track_id, raw_confidence

class IdentityFusion:

```

```

    bearing_weight, ratio_weight, motion_gate_strength
    per_label_filters: map of label -> ScalarKalmanFilter    # smooths confidence over
time

    function update(labels, detections, bearing_cost, ratio_cost, approach_estimates):
        blended_cost = blend(bearing_cost, ratio_cost, bearing_weight, ratio_weight)
        # if one cue is missing (None), fall back to using only the other, unweighted

        row_idx, col_idx = hungarian_assignment(blended_cost)
        # one detection per candidate label, minimizing total cost

        results = {}
        for each (label, detection) matched by the assignment:
            raw_confidence = 1 - blended_cost[label][detection]
            gated_confidence = apply_motion_gate(raw_confidence,

approach_estimates.get(detection.track_id))

            filt = per_label_filters.get(label) or new ScalarKalmanFilter
            filt.predict(dt=1); filt.update(gated_confidence)
            fused = filt.value

            results[label] = CandidateConfidence(label, fused, detection.track_id,
raw_confidence)
        return results

    function apply_motion_gate(confidence, estimate):
        if estimate is None: return confidence
        return confidence + motion_gate_strength if estimate.approaching
            else confidence - motion_gate_strength

```

bronchotrack/fusion/association.py

End-to-end continuous re-evaluation: re-runs the fused match every frame against whatever is currently visible, and commits a branch transition only after a consecutive-frame confidence streak.

```

class AssociationResult:
    fields: location, committed_this_frame, transition_sanity, candidates

function detection_size(tracklet):
    same as motion_model's diameter proxy (box height)

class FusionAssociation:
    graph, commit_threshold, commit_frames
    current_location, has_committed_once = False
    commit_streak: map of label -> consecutive-frame count
    motion_model = ApproachMotionModel()
    sanity_checker = TransitSanityChecker()
    fusion_by_location: map of location -> IdentityFusion    # one fusion instance per
branch point
    eligibility_fn: always returns True (no tracker dropout in this design)

    function process_frame(tracklets, frame_idx):
        approach_estimates = motion_model.update(tracklets, frame_idx)
        motion_model.prune(live track ids)

        if not has_committed_once:
            return try_bootstrap(tracklets, frame_idx)

        reference = pick_reference(tracklets)

```

```

# the tracklet currently labeled as occupying current_location
if reference is None:
    return AssociationResult(current_location, committed=False,
                             sanity="unknown", candidates={})

children = graph's children of current_location
fusion = get_fusion(current_location) # lazily created IdentityFusion per
location

bearing_cost = bearing_cost_matrix(graph, current_location, children,
                                    tracklets, detection_point(reference))
ratio_cost = ratio_cost_matrix(graph, current_location, children, reference,
                                tracklets, virtual_advance_mm)
candidates = fusion.update(children, tracklets, bearing_cost, ratio_cost,
                           approach_estimates)

committed, sanity_verdict = try_commit(candidates, tracklets, frame_idx)
return AssociationResult(current_location, committed, sanity_verdict or
"unknown",
                        candidates)

function pick_reference(tracklets):
    return the tracklet whose .label == current_location, if any, else None

function try_bootstrap(tracklets, frame_idx):
    # special one-time case: at the very first bifurcation there is no
    # pre-existing occupant, so both visible lumens are labeled from
    # bearing alone in a single shot
    children = graph's children of the graph root
    match each visible tracklet to the nearest child bearing
    (bearing_cost_matrix only, no ratio/motion yet)
    label every matched tracklet accordingly
    current_location = label of the larger / most prominent matched tracklet
    has_committed_once = True
    return AssociationResult(current_location, committed=True,
                             sanity="unknown", candidates={})

function try_commit(candidates, tracklets, frame_idx):
    for label, cand in candidates:
        if cand.fused_confidence >= commit_threshold:
            commit_streak[label] += 1
        else:
            commit_streak[label] = 0 # reset immediately, not decayed

    best_label = candidate with highest streak that has reached commit_frames
    if no such label: return (False, None)

    matched_tracklet = tracklet with track_id ==
candidates[best_label].matched_track_id
    arc_length_mm = graph distance from current_location to best_label
    tau = motion_model.tau_frames_for(matched_tracklet.track_id)
    sanity = sanity_checker.check(tau, arc_length_mm) if tau is not None else
"unknown"
    if sanity resolves and location actually changes:
        sanity_checker.record_transit(best_label,
frames_spent=commit_streak[best_label],
                                arc_length_mm)

    current_location = best_label
    reset commit_streak for all labels
    update_gallery(matched_tracklet, best_label)
    return (True, sanity)

```

```

function update_gallery(tracklet, label):
    tracklet.label = label      # so pick_reference finds it next frame

function generation(label):
    return graph node's generation depth (used for HUD / logging)

```

bronchotrack/fusion/pipeline.py

Drives detector, tracker, and association together frame by frame, writes the overlay video and the JSON log.

```

class FrameResult:
    fields: frame_idx, location, committed_this_frame, candidates, tracklets

class FusionPipeline:
    detector, tracker, association: FusionAssociation, graph

    function process_frame(frame_bgr, frame_idx):
        detections = detector.infer(frame_bgr)
        active_tracklets = tracker.update(detections, frame_idx,
                                          eligibility_fn=association.eligibility_fn)
        result = association.process_frame(active_tracklets, frame_idx)
        return frame_log_entry(frame_idx, result, active_tracklets)

    function run_on_video(video_path, out_video_path, out_json_path):
        open video reader, video writer
        log = []
        for each frame, frame_idx in video:
            result = process_frame(frame, frame_idx)
            overlay_frame = viz.draw_overlay(frame, result, graph)
            write overlay_frame to out_video
            log.append(result)
        write_json_log(log, out_json_path)
        close reader/writer

    function frame_log_entry(frame_idx, result, tracklets):
        return dict with frame_idx, current location, committed flag,
            per-candidate confidences, per-tracklet box/label info

    function write_json_log(log, path):
        serialize log list to JSON file

```

bronchotrack/fusion/viz.py

Draws the live overlay: tracked lumens plus a HUD panel listing every currently scored candidate child branch (not fabricated screen positions, since no camera calibration exists in this pipeline).

```

function draw_overlay(frame_bgr, result, graph):
    out = copy of frame_bgr
    for each tracklet currently visible:
        draw a dot / box at its position
        # no full mask HUD placement -- no camera calibration exists to place
        # candidate branches at real screen positions

    draw_candidate_hud(out, result)      # side panel listing every scored candidate
    child
    draw_current_location_label_near_top_of_frame

```

```

    return out

function draw_candidate_hud(out, result):
    panel_x, panel_y = fixed HUD corner
    for each (label, candidate_confidence) in result.candidates, sorted by confidence
    descending:
        draw label name
        draw a horizontal bar sized proportional to fused_confidence
        draw ETA text (from tau_frames if available, else a placeholder dash)
        advance panel_y for next row

```

bronchotrack/fusion/cli.py

Command-line entry point: parses arguments, wires up the detector, tracker, association, and pipeline, then runs the video.

```

function build_arg_parser():
    args: --video, --graph, --weights, --conf-threshold, --out-video, --out-json,
          --commit-threshold, --commit-frames, --show-graph (mirrors paper_exact's
CLI)
    return parser

function main(argv):
    args = build_arg_parser().parse_args(argv)
    graph = AirwayGraph.from_json(args.graph)
    detector = load detector from args.weights
    tracker = MultiLumenTracker()
    association = FusionAssociation(graph, commit_threshold=args.commit_threshold,
                                   commit_frames=args.commit_frames)
    pipeline = FusionPipeline(detector, tracker, association, graph)

    pipeline.run_on_video(args.video, args.out_video, args.out_json)
    print("Processed <N> frames... Final location: <label> (generation <g>)")

```

bronchotrack/fusion/__init__.py

Package entry point: exports the public classes and documents the three fused models and the deliberate scope reductions for this first version.

```

# Public exports:
from .association import AssociationResult, FusionAssociation
from .kalman_fusion import CandidateConfidence, IdentityFusion
from .motion_model import ApproachEstimate, ApproachMotionModel, TransitSanityChecker
from .pipeline import FrameResult, FusionPipeline

# __all__ = [FusionAssociation, AssociationResult, IdentityFusion, CandidateConfidence,
#            ApproachMotionModel, ApproachEstimate, TransitSanityChecker,
#            FusionPipeline, FrameResult]

# No executable logic beyond imports/exports -- this module is the package's
# documented overview:
# 1. motion_model.ApproachMotionModel -- diameter-growth, scale-free tau
# 2. bronchotrack_id -- point-based bearing matching (segmentation center)
# 3. ratio_id -- whole-mask ratio method (area + boundary-gap, not points)
# kalman_fusion.IdentityFusion blends (1) and (2) into one Hungarian
# assignment per frame; association.FusionAssociation re-runs this every
# frame (continuous re-evaluation) and commits only after a consecutive
# confidence streak.

```

```
#
# Deliberate scope reductions for this first version (flagged, not hidden):
# - no roll-angle correction in bronchotrack_id (plain unrotated bearing)
# - no multi-generation "ambiguous cohort" localization smoothing
# - no tracker eligibility dropout (eligibility_fn always True)
```

tests/test_fusion_smoke.py

Smoke and regression tests for the fusion package: geometry primitives, motion model behavior, fusion blending, commit hysteresis, and a no-ML end-to-end run. 18 tests total.

```
helper _rect_mask(cx, cy, half): builds a square polygon mask centered at (cx, cy)
helper _mk_tracklet(track_id, cx, cy, h, mask, label, hits): builds a minimal test
Tracklet

test_child_apparent_equivalent_diameter_matches_ellipse_area_formula:
    build a straight branch and a 60-degree-off-axis branch off a common root
    assert straight branch reads ~true diameter; off-axis area-based estimate is
    larger than the off-axis longest-diameter estimate (matches the derived
    cos vs sqrt(cos) foreshortening formulas exactly)

test_mask_boundary_gap_and_circle_boundary_gap_exact:
    two rectangles separated by a known horizontal gap -> mask_boundary_gap
    returns that exact gap
    two circles at known centers/radii -> circle_boundary_gap returns the exact
    gap; overlapping circles clamp to MIN_DIST

test_ratio_method_disambiguates_where_bearing_alone_cannot:
    construct two candidate branches with identical projected bearing but
    different radius, two detections at the identical image point but
    different mask size
    assert bearing_cost_matrix is completely flat (no discriminating power)
    assert ratio_cost_matrix correctly favors the size-matched pairing

test_ratio_cost_matrix_falls_back_gracefully_without_radius_data:
    graph with no radius data -> ratio_cost_matrix returns None

test_ratio_cost_matrix_handles_no_other_detections_this_frame:
    empty detections list -> ratio_cost_matrix returns None cleanly,
    no NaN / RuntimeWarning

test_bearing_cost_matrix_hungarian_correct_assignment:
    two candidates at right-angle bearings, two detections offset to match each
    assert cost matrix is exactly [[0,1],[1,0]]

test_motion_model_tau_matches_hand_calculation:
    feed linearly growing diameter for 40 frames
    assert estimate.approaching is True and tau_frames is approx
    diameter / growth_rate

test_motion_model_not_approaching_when_shrinking:
    feed shrinking diameter -> approaching is False, tau_frames is None

test_motion_model_prune_drops_stale_filters:
    update with one tracklet, then prune with an empty live set -> filter removed

test_transit_sanity_checker_plausibility:
    no history -> "unknown"
    record two consistent transits -> median speed computed correctly
    a consistent candidate -> "plausible"; a wildly different one -> "implausible"
```

```

test_identity_fusion_blends_cues_and_motion_gates:
    two candidates with identical bearing+ratio cost, but only one detection
    is "approaching"
    assert the approaching candidate's fused_confidence ends up higher

test_identity_fusion_falls_back_to_single_cue_when_other_missing:
    bearing_cost = None -> fusion uses ratio_cost alone, unweighted

test_fusion_association_root_bootstrap_labels_both_lumens_at_once:
    two lumens visible at the very first frame, no prior occupant
    assert both get labeled correctly in one frame and the larger becomes
    current_location

test_fusion_association_commits_after_consecutive_frames_not_one_lucky_frame:
    run commit_frames-1 strong frames -> not yet committed
    run one more strong frame -> commits to the correct child

test_fusion_association_streak_resets_on_interrupting_bad_frame:
    4 clearing frames, 1 frame below threshold, 4 more clearing frames
    (commit_frames=5)
    assert it never commits, because the streak resets on the bad frame
    instead of just pausing

test_fusion_association_continuous_reevaluation_survives_reference_loss:
    bootstrap, then a frame with zero tracklets
    assert no crash and current_location is unchanged

test_scalar_kalman_filter_promoted_and_reexported_identically:
    assert bronchotrack.scalar_kalman.ScalarKalmanFilter is the same object as
    bronchotrack.paper_exact.scalar_kalman.ScalarKalmanFilter (re-export,
    not a copy)

test_full_fusion_pipeline_smoke_no_ml:
    run tracker + FusionAssociation together on synthetic carina-region
    detections (no real ML)
    assert both main bronchi get bootstrapped and labeled correctly

_run_all(): runs every test above in sequence, prints PASS/FAIL per test and
a final N/M summary

```